From Classical to Quantum Computing: Working Together on Grand Challenges

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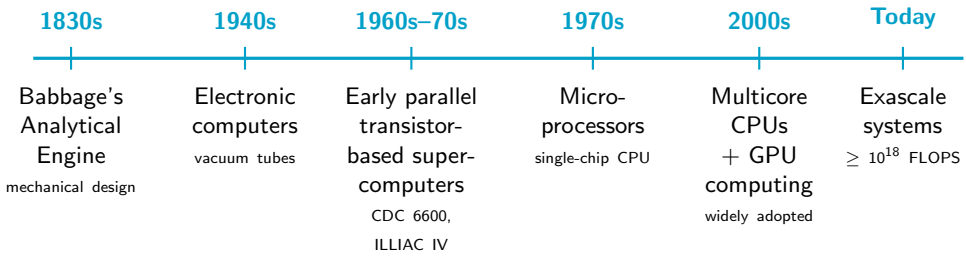
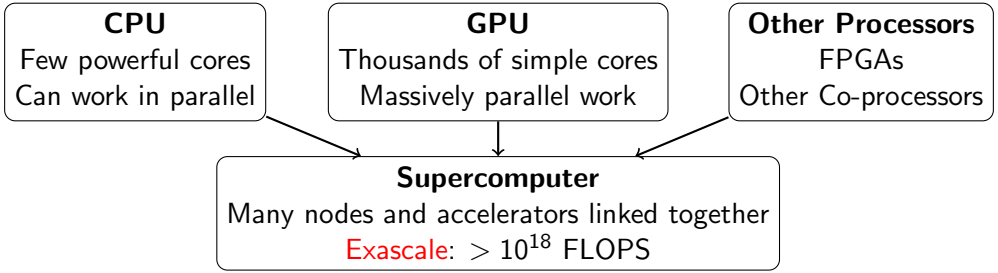
From Mechanical Ideas to Exascale Computing

Computation evolved from mechanical designs into electronic machines capable of performing more than 10^{18} floating point operations per second.



The goal remained the same: automating computation. What changed was the technology: from mechanical motion to electronic circuits, early parallel machines, microprocessors, accelerators, and exascale supercomputers.

Present Scenario



EuroHPC JU: Our Supercomputers

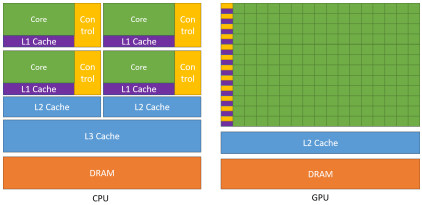
EuroHPC JU supercomputers: Overview of EuroHPC systems, locations, and capabilities.

TOP500

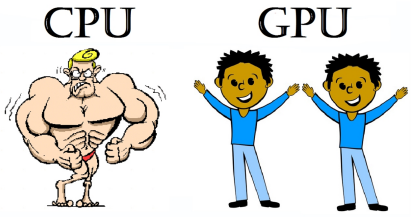
TOP500 ranking: Global ranking of the world's most powerful supercomputers.

Important differences between CPU and GPU

- GPU has many cores compared to CPU
- But on the other hand, the CPU's frequency is higher than the GPU. That makes the CPU faster per core than the GPU.
  - Intel® Core™ i7-10700K Processor base frequency is **3.80 GHz**, whereas, Nvidia Ampere has **0.765 GHz**



Source:Nvidia: CUDA programming



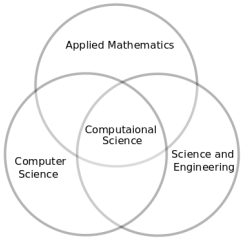
Supercomputers from Europe

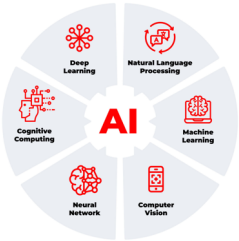
Figure: BSC–Barcelona

Figure: JUPITER–Jülich, Germany

Figure: LUMI–Finland

Scientific Computing - System of Linear Equations

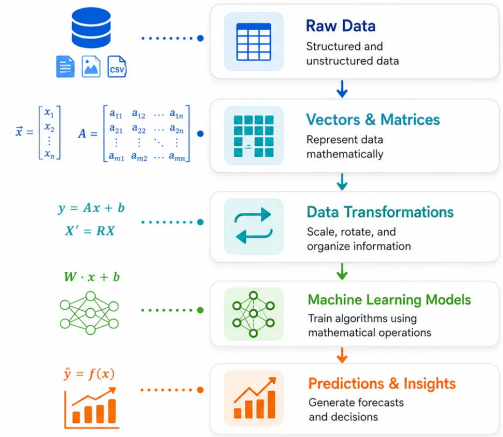


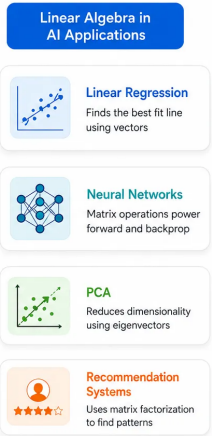


$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}, \quad \boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

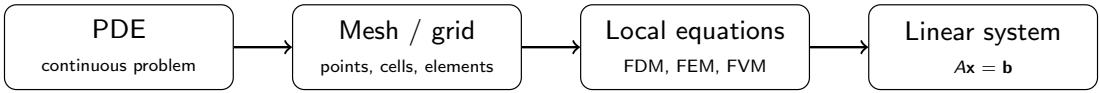
- Many problems in science and engineering are described by differential equations, especially partial differential equations (PDEs).
- These PDEs describe how physical quantities change in space and time.
  - Examples include heat transfer, fluid flow, wave propagation, weather forecasting, and structural mechanics.
- To solve PDEs on a computer, the continuous physical domain must be discretized into a finite number of points, cells, or elements.
- These systems are then solved using direct or iterative solvers to determine the unknown variables.
- Similarly, many artificial intelligence applications involve large matrix and vector computations.

Why AI and ML Need Linear Algebra





Science and Engineering: Discretization Leads to Linear Systems



Finite difference example

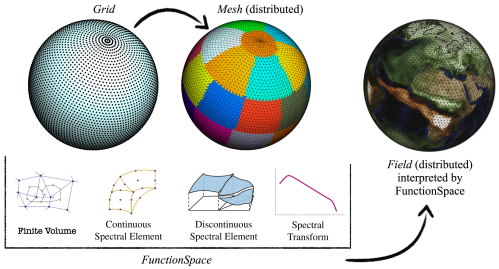
$$-\frac{d^2u}{dx^2} = f(x) \qquad \Longrightarrow \qquad \frac{-u_{i-1} + 2u_i - u_{i+1}}{h^2} = f_i$$

$$\underbrace{\frac{1}{h^2}\begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}}_{A\text{: coefficients from the stencil}} \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}}_{\mathbf{x}\text{: unknown values}} = \underbrace{\begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}}_{\mathbf{b}\text{: known data}}$$

Each grid point gives one equation; each equation becomes one row of A.

Applications and Examples

- In weather forecasting, the domain size is discretized by numerical methods. For example, 786 thousand cores from the IBM Blue Gene/Q Mira system at Argonne National Laboratory are needed to simulate a resolution of 3 km (1.8 billion cells) using spectral element methods (one kind of numerical method)[1].



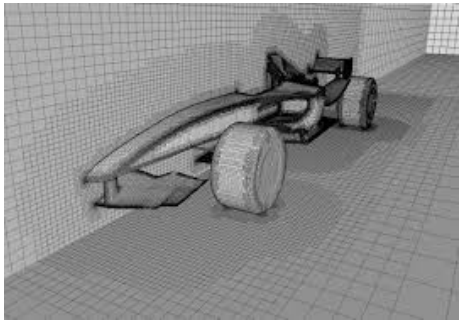
Mesh and simulation settings [2]

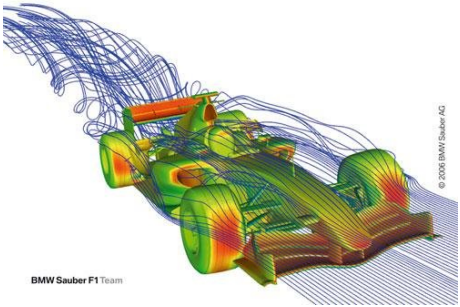
[1] HPC for Weather Forecasting, John Michalakes

[2] A library for numerical weather prediction and climate modeling, Willem Deconinck, et al.

Applications and Examples

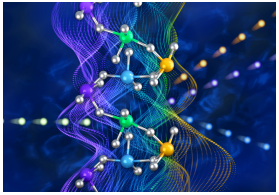
- CERFACS in France used up to 250 000 cores with grids of 2 to 4 billion cells (using Large Eddy Simulation - one kind of numerical methods strategy) for computational fluid dynamics application.

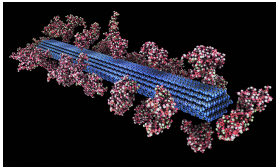




Applications and Examples

- **Materials science:** design new materials with desired properties.
- **Biology:** understand proteins and support drug discovery.
- **Engineering:** simulate fluids, heat transfer, and structures to improve designs.



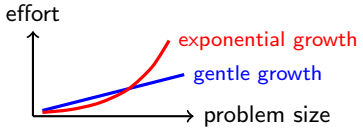


Limitations of Classical Computers

- Classical computers and supercomputers are powerful tools for scientific computing, especially for solving PDEs with numerical methods.
- However, CPU performance is increasingly limited by three walls:
  - **Memory wall:** data movement is slower than computation.
  - **Power wall:** higher frequency increases heat and power consumption.
  - **ILP wall:** instruction dependencies limit parallel execution.
- These limits motivate alternative and complementary computing approaches, such as GPUs, accelerators, and **QUANTUM COMPUTING**.

Some Problems Push Back

Some problems grow exponentially harder as they get bigger. More processors alone are not enough.



Example: quantum systems

Simulating molecules is difficult because the number of possible quantum states can grow exponentially with system size.

- **Quantum chemistry:** molecular states can grow exponentially.
- **Drug discovery:** many possible molecule–protein interactions.
- **Materials science:** electronic states grow rapidly with system size.
- **Optimization:** possible solutions can grow exponentially.

A natural idea

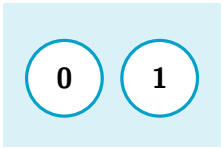
If nature itself is quantum,

perhaps a quantum machine is the natural tool to reason about it.

(an idea often credited to Richard Feynman, 1982)

Bit versus qubit

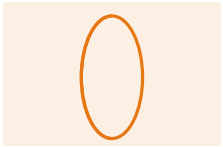
Classical bit



exactly one value

A coin lying flat: heads or tails.

Quantum qubit



0 and 1 together

A coin spinning: both at once, until it lands.

The spinning-coin analogy

- A spinning coin is neither heads nor tails — a blend of both. (superposition)
- **1 coin** $\rightarrow$ 2 states. **2** $\rightarrow$ 4. **10** $\rightarrow$ 1,024. $n \rightarrow 2^n$.
- Qubits can be linked so their outcomes correlate. (entanglement — Einstein's "spooky action")

The promise

A modest number of qubits represents an enormous space of possibilities at once.

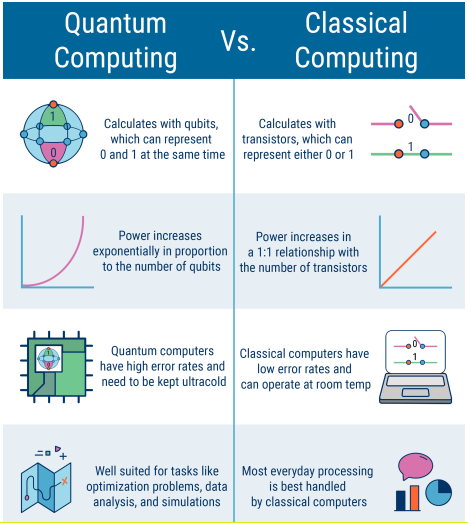
But thers is a catch

Take-away

Quantum computers are not just faster classical computers. They compute in a fundamentally different way — powerful for some problems, irrelevant for most.

When you measure, the spinning coin lands on one answer.

- We cannot read out all 2^n possibilities.
- A quantum algorithm arranges interference so the right answer is most likely to appear.

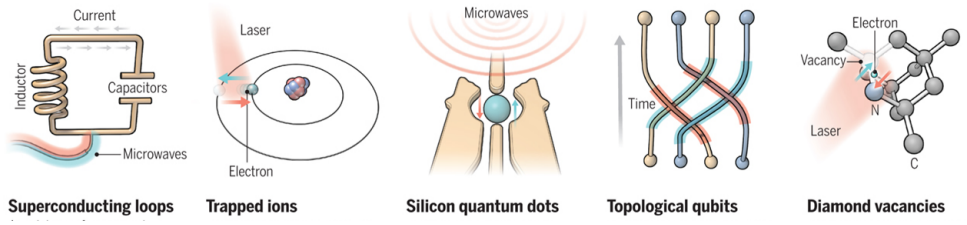


The hardware, in one sentence

Many ways to build qubits — superconducting circuits, trapped ions, neutral atoms, photons —

but all face the same challenge:

keeping quantum information stable and reducing errors.



Noise, errors, and the early stage

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Decoherence

A spinning coin is fragile: heat, vibration, or stray fields make it "fall over" and lose its state.

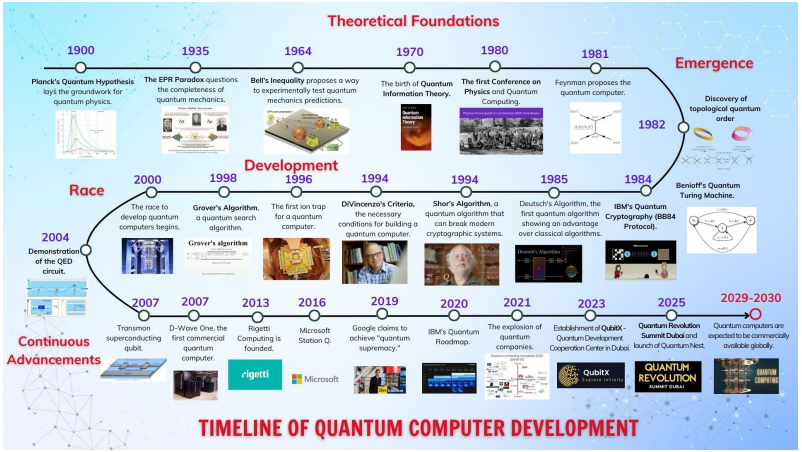
Errors

Operations are imperfect; errors accumulate, limiting how long a computation can run.

- Today's machines are noisy and small — the "NISQ" era.
- Error correction spreads one reliable "logical" qubit across many physical ones.
- Works in principle — but needs far more, better qubits than we have.

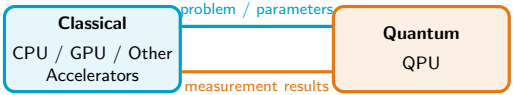
An honest status check

Quantum computing is real and progressing fast, but still early.

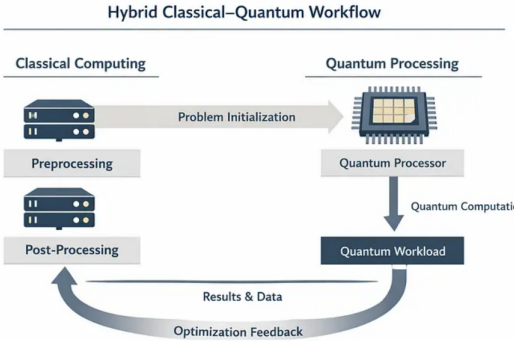


Not a replacement — a partnership

Quantum processors are specialists. They sit alongside classical computing (CPUs), each doing what it does best.



The two loop back and forth until the problem is solved.



Application 1 — Materials & drug discovery

- Molecules and materials are governed by quantum mechanics.
- Classical simulation hits a wall for strongly interacting electrons.
- Quantum hardware could model molecules natively.

Could mean: better batteries, catalysts, fertilizers; faster drug screening.

The hybrid split

Quantum: the electronic-structure core.

Classical: geometry, sampling, ranking candidates.

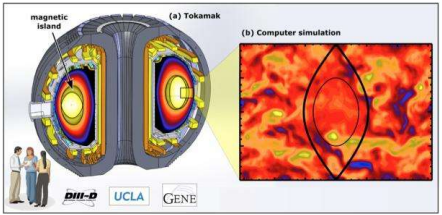
Application 2 — Physics simulations

- Plasma simulation: involves turbulence, which requires enormous computational power.
- Quantum routines could accelerate selected sub-problems, not whole simulations.

The hybrid split

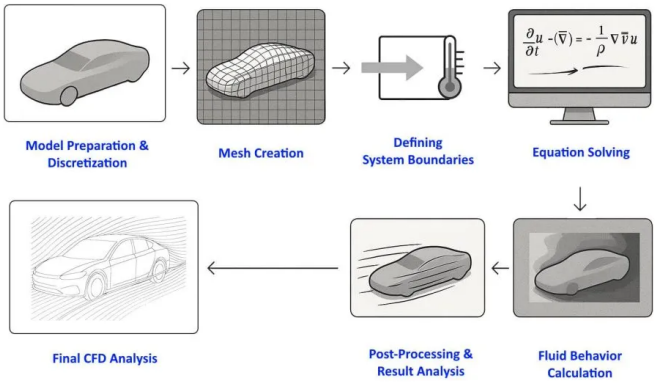
Classical HPC: the full simulation and visualization.

Quantum: targeted kernels that are bottlenecks classically.



Plasma Simulation.

Application 2 — Physics simulations



- Airbus and Rolls Royce are looking into utilizing quantum computers for airplane and turbojet engines.

Application 3 — Optimization & machine learning

- Scheduling, routing, portfolios: vast spaces of possible choices.
- Quantum methods may explore certain landscapes differently.
- In ML, quantum models could enrich how data is represented.

Reality check: advantages here are problem-specific and still being established.

The hybrid split

Classical: train, optimize, decide.

Quantum: propose or evaluate hard candidates in the loop.

Quantum computing in Europe — available now

Europe's strategy mirrors this talk: quantum integrated with classical HPC.

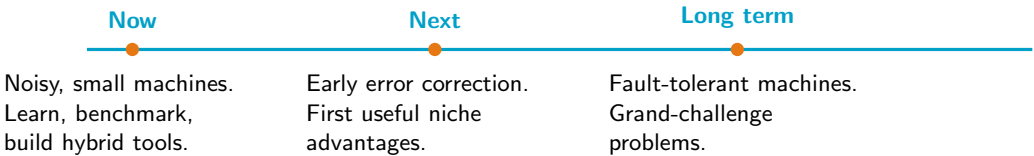
- **EuroHPC JU** is deploying six quantum systems, each coupled to a Tier-0 supercomputer — superconducting, trapped-ion, photonic, and neutral-atom platforms.
- **Superconducting:** Euro-Q-Exa (LRZ, Germany) — 54 qubits now, 150 by end of 2026. IQM (Finland/Germany) builds the hardware.
- **Trapped ion:** AQT (Innsbruck, Austria) — 12-qubit IBEX on the cloud, first quantum hardware for EU customers on AWS Braket.
- **Neutral atom:** SOL (Bologna, Italy) — neutral-atom quantum simulator.
- **Annealing:** D-Wave being coupled to JUPITER (Jülich, Germany), Europe's first exascale machine.

Hybrid systems in practice — happening now

The recurring pattern: the QPU becomes just another accelerator in the HPC job queue, next to CPUs and GPUs.

- **Europe (LRZ, Germany):** IQM quantum computers run as a Slurm node inside the supercomputer — quantum jobs scheduled alongside CPU/GPU jobs, using the same workload manager HPC already relies on.
- **US national labs:** DOE integrates IBM, Rigetti, and IonQ hardware at Oak Ridge and Argonne; quantum subroutines embedded inside classical simulation pipelines (Qiskit Runtime, Braket Hybrid Jobs).
- **Exascale link:** optimization algorithms (QAOA) run in hybrid workflows tied to Frontier, connecting quantum kernels to a classical exascale machine.
- **Programming model:** NVIDIA CUDA-Q lets one program co-target CPU + GPU + QPU together.

A realistic roadmap



Progress is incremental — and hybrid systems deliver value at every step.

Conclusion

- Classical computing evolved relentlessly — and powers modern life.
- A few problems remain out of reach for any classical machine.
- Quantum computing is a fundamentally different way to compute — powerful for select problems.
- It is early: noise and errors are the central challenge.
- The future is quantum with classical — hybrid systems for grand challenges in materials, biology, physics, engineering, and AI.

Working together on the problems that matter most.

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Thank you

Questions?